Two Machine-Checked Conditional Routes to the Riemann Hypothesis

A Formalized Fekete Criterion, Effective Positivity Targets, and an Exact-Ready Receptor Architecture

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Abstract

We do not prove the Riemann Hypothesis. We present a Lean 4 / Mathlib formalization (76 modules, ≈ 29,000 lines, 122 audited theorems depending only on Lean's three foundational axioms, with no `sorry` and no project axioms) that reduces RH to explicitly named open positivity inputs along two independent, *exact-ready* routes, and that isolates—by proof and by machine-checked counterexample—precisely which new mathematics remains.

The formal core is a machine-checked Fekete-type total positivity criterion: for the lower-triangular Toeplitz matrix of the signed Taylor coefficients of Riemann's Ξ-function, *strict positivity of all contiguous minors implies full total positivity.* The proof mechanizes a generic Grassmann syzygy and a three-term gap-shrinking identity valid at every matrix order, an all-orders double induction (on order and row dispersion), a dominance dichotomy whose vanishing half is a permutation-pigeonhole argument, and a Toeplitz reversal symmetry. Combined with the classical scaffolding (Aissen–Schoenberg–Whitney/Edrei, Laguerre–Pólya closure, Pólya–Jensen), this yields a minimal named conditional reduction: RH follows from the strict contiguous ladder alone, with no initial-column criterion and no kernel representation.

We further formalize a bounded-degree hyperbolic-limit theorem (a finite-dimensional Hurwitz-type statement with a fully elementary proof), a receptor architecture that lets each future ingredient—an explicit remainder bound, a certified prefix, or the general same-sign Schur–Szegő composition theorem—close its route by a single `exact`, and a machine-checked counterexample showing that the known second- and third-order Turán inequalities for the Ξ coefficients cannot, at the sequence level, close even the order-3 rung of the ladder. We conclude with a research map of the new mathematics the architecture is ready to receive.

Keywords: Riemann hypothesis; Pólya frequency sequences; Toeplitz total positivity; Jensen polynomials; formal mathematics; Lean 4.

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1 Introduction

The Riemann Hypothesis (RH) [1] admits a classical reformulation in the language of total positivity: writing $\Xi(t) = \sum_{n\geq 0} b_n t^{2n}$ for the Taylor expansion of Riemann's Ξ-function and $\mu_n = (-1)^n b_n$ for its signed coefficients, RH is equivalent (through the Laguerre–Pólya theory and the Aissen–Schoenberg–Whitney–Edrei characterization [4, 5]) to the assertion that (μ_n) is a *Pólya frequency sequence*: every minor of the one-sided infinite Toeplitz matrix

$$T(i,j) \;=\; \begin{cases} \mu_{i-j}, & j \leq i, \\ 0, & j > i, \end{cases}$$

is nonnegative. The order-1 and order-2 fragments of this condition are classical theorems—positivity of the μ_n follows from Pólya's integral representation [2], and the order-2 (Turán) inequalities were proved by Csordas, Norfolk and Varga [10], resolving a problem posed by Pólya [3]. Everything at order ≥ 3 is open and is, in aggregate, equivalent to RH.

This paper reports a formalization program, carried out in Lean 4 over Mathlib [23, 24], whose purpose is *not* to prove RH but to do three things that we believe are individually useful and jointly new:

1. **Prove, with kernel-checked proofs, every step of the reduction that is genuinely provable today**, so that the open mathematics is isolated into a small number of explicitly named hypotheses (§3). The formal centerpiece is an all-orders Fekete-type criterion (§5): strict positivity of the *contiguous* Toeplitz minors implies the full Pólya frequency condition.
2. **Assemble two independent, exact-ready conditional routes to RH** (§6–7), each terminating in a Lean theorem whose hypotheses are named propositions:
   - the kernel/PF route `riemannHypothesis_of_effectiveCertificates`, whose single open input `KernelEffectiveCertificates` asks, for each matrix order, for a certified finite prefix together with an explicit remainder-bound tail; and
   - the Jensen/HPSS route `riemannHypothesis_of_hpss_route`, whose open inputs are the Pólya–Jensen bridge, the general same-sign Hermite–Poulain/Schur–Szegő composition theorem, and per-window hyperbolicity of the coefficient polynomials.
3. **Prune dead branches by machine-checked falsification** (§8): we exhibit a kernel-checked counterexample showing that the known local Turán data (orders 2 and 3) cannot close even the order-3 contiguous rung at the sequence level, and we record why a recent certified PF_5 failure for the continuous de Bruijn–Newman kernel [19] forces any proof to target the discrete moment sequence directly.

Figure 1 summarizes the resulting architecture: each open input is a single named proposition, and every arrow below it is a kernel-checked theorem or a named classical input.

Claims and non-claims. No theorem in this paper or in the underlying repository proves RH, and none assumes it: every RH-equivalent or RH-adjacent statement is carried as a named *hypothesis* (a Lean `Prop`), never as an axiom. An automated audit (§3) verifies that all 122 audited theorems depend on nothing beyond Lean's three foundational axioms (`propext`, `Classical.choice`, `Quot.sound`). In particular, we prove (Corollary 5.6) that our main positivity hypothesis is *at least RH-strength*, which is exactly why it must remain a hypothesis. What the formalization contributes is the complete, kernel-checked closure of

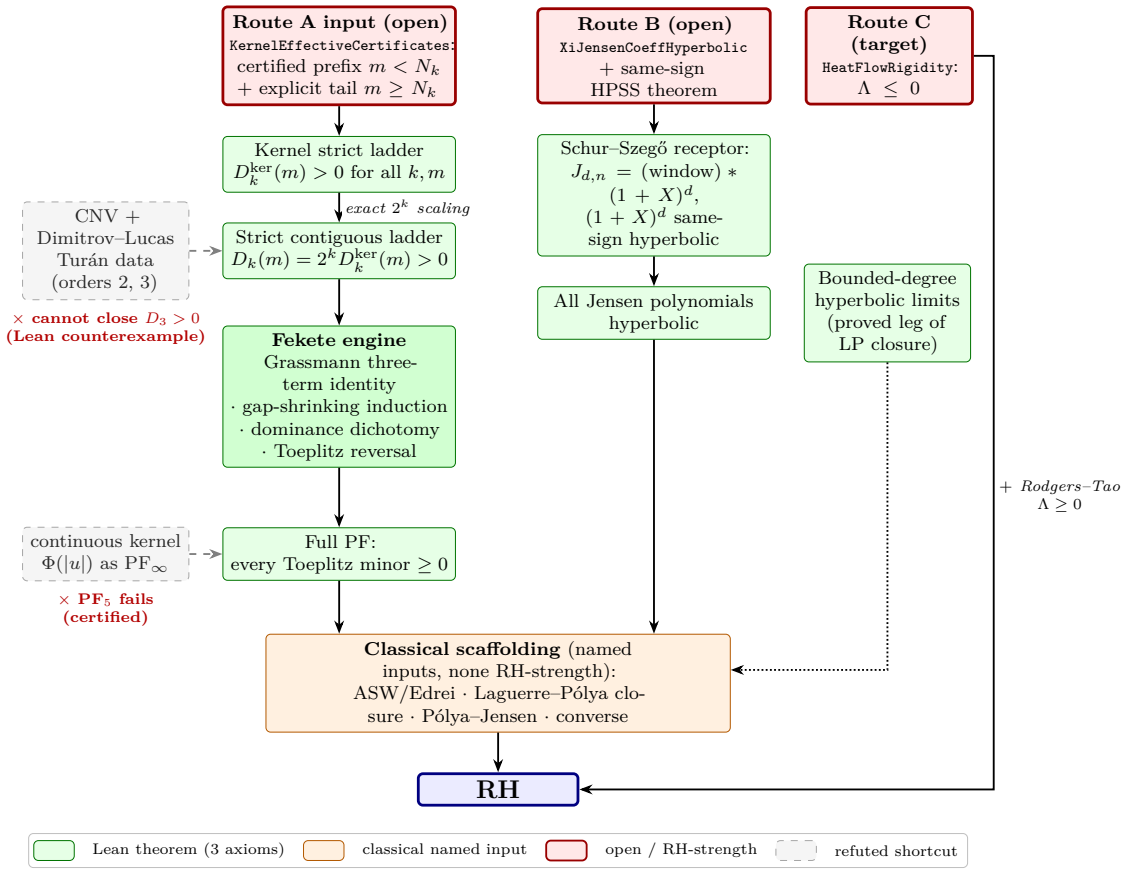


Figure 1: Architecture of the two conditional routes. Green: Lean theorems whose axiom footprint is Lean's three foundational axioms. Amber: classical inputs carried as named hypotheses. Rose: the open inputs that close each route by a single `exact`; the strict ladder they feed is at least RH-strength (Corollary 5.6). Grey: shortcuts ruled out by a machine-checked counterexample (§8) or by the certified PF_5 failure of the continuous kernel [19].

everything between that hypothesis and RH—a closure which, to our knowledge, had not previously been carried out in full even informally at this level of granularity.

2 Background and conventions

We use the completed zeta function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and its real form $\Xi(t) = \xi(\frac{1}{2} + it)$, an even entire function of order one, real and positive at $t = 0$. In the formalization the coefficient sequence is *defined* derivative-first,

$$b_n \;=\; \texttt{XiCoeff}\, n \;=\; \frac{\Xi^{(2n)}(0)}{(2n)!}, \qquad \mu_n \;=\; (-1)^n b_n,$$

so that all downstream theorems are anchored to a concrete real sequence; the analytic identification of (b_n) with the Taylor coefficients and Pólya's kernel representation $b_n = (-1)^n\, 2M_n/(2n)!$ with $M_n = \int_0^\infty \Phi(u)\, u^{2n}\, du$ and $\Phi > 0$ [2] are named classical inputs, not silent assumptions.

For $k, m \in \mathbb{N}$ the *contiguous minor of order k at offset m* is

$$D_k(m) \;=\; \det\big[\, T(m+i,\, j)\,\big]_{0 \le i,j < k},$$

the determinant of the $k \times k$ block whose rows are consecutive and whose columns are the initial interval $\{0, \ldots, k-1\}$ (`XiToeplitzContigMinor`). Three positivity conditions on the family of *all* minors will occur:

- *Contiguous, strict:* `XiContigToeplitzStrictPositive`, meaning $\forall k, m,\ D_k(m) > 0$.
- *Initial-column:* `XiInitialColumnMinorTotalPositive`, meaning arbitrary rows with columns $0, \ldots, k-1$ and determinant ≥ 0.
- *Full PF:* `XiToeplitzTotalPositive`, meaning arbitrary rows and columns with determinant ≥ 0.

The classical scaffolding is bundled as a structure `ClassicalToeplitzScaffolding` with four fields: the ASW/Edrei bridge from full PF to real-rooted finite approximants of Ξ [4, 5], the Laguerre–Pólya closure bridge, the Pólya–Jensen bridge (RH $\iff$ hyperbolicity of all Jensen polynomials [3]), and the classical converse (RH $\Rightarrow$ full PF). Each is a proved theorem of classical analysis whose formalization is a separate, ongoing program (§9); none is RH-strength.

3 Formal methodology

The repository comprises 71 Lean modules ($\approx$ 28,000 lines) over Mathlib [23]. Three disciplines are enforced mechanically.

Named hypotheses, never axioms. Open mathematics enters only as a named `Prop` passed explicitly to conditional theorems. A source scanner rejects `sorry`, `admit`, and project-level `axiom` declarations in the entire library.

Semantic axiom audit. A dedicated module runs `#printaxioms` on the 122 headline theorems; a CI script fails unless every one reports exactly

$$[\texttt{propext},\ \texttt{Classical.choice},\ \texttt{Quot.sound}].$$

This guards against silent placeholder reintroduction: assuming an RH-equivalent statement anywhere in a proof would surface immediately.

Falsification before formalization. Numerical claims are treated as falsification evidence, never proof. Interval-certified scans (exact rational interval arithmetic over decimal enclosures of the first 121 coefficients, indexed 0 through 120) test each named positivity target before formal work begins; certified negative signs would kill a route, and one such sequence-level route-kill is itself formalized (§8).

Differential numerical replication. The bounded minor scan is implemented independently in Python, JavaScript, and TypeScript. Python and TypeScript use exact interval arithmetic under the stated half-ULP decimal model; JavaScript uses exact midpoint `BigInt` arithmetic and a fraction-free Bareiss determinant. Positive, zero, and negative controls, the coefficient-file SHA-256 digest, and full sign-stream hashes are compared automatically. Agreement strengthens reproducibility of the finite scan but does not certify the upstream coefficient extraction or any infinite claim.

4 Synthesis of the formalization program and the literature

The repository contains more than one proposed route to RH. A publishable account must distinguish the common analytic object, the unconditional Lean algebra, the conditional interfaces, and the exploratory programs that were rejected or demoted. This section records that synthesis so that the paper is not read as claiming that an equivalent reformulation is itself progress on RH.

4.1 The corrected object and the common spine

An early version of the development used the meromorphic completed-zeta variant Λ_0. That is not the correct object for a Pólya–Jensen or Pólya-frequency argument: its critical-line zeros do not coincide with the non-trivial zeros of ζ. The current formalization instead defines

$$\xi(s) := \frac{s(s-1)}{2}\,\pi^{-s/2}\Gamma(s/2)\zeta(s), \qquad \Xi(t) := \xi\Big(\tfrac{1}{2} + it\Big)\,, \tag{1}$$

and proves the zero-correspondence theorem `Xi_eq_zero_iff_riemannZeta`. Every coefficient, Jensen, Toeplitz, and moment statement in this paper is downstream of this corrected spine. The correction is a soundness repair, not a claim of a new theorem about ζ.

4.2 What is retained, and what is demoted

Table 1 gives the status of every major route developed in the project. "Proved" means proved in Lean from the ordinary foundational axioms; it does not mean that an external analytic hypothesis has been discharged.

Table 1: Synthesis of the formalized RH program. The final column identifies the obstruction that prevents an unconditional proof.

Layer	**Lean contribution**	**Remaining qualification**
Same-height fibers	Centroid, energy, and cardinality reductions	RH-equivalent restatements; no repulsion theorem
Li / Hilbert–Pólya	Symmetry and conditional witness packages	Li data and operator existence are external inputs
3–4–1	Zero-free boundary bridge and certificate barrier	Classical near-$\operatorname{Re} s = 1$ result; cannot reach $1/2$
Fekete–Toeplitz	Grassmann identity, all-order gap shrinking, full-PF capstone	Strict contiguous ladder is at least RH-strength
Jensen / HPSS	Hyperbolic-limit theorem and exact receptor	All-degree/all-shift window input and general same-sign composition remain open
Heat flow	Named third-route interface	Requires a new rigidity theorem proving $\Lambda \leq 0$

The same-height and Li/Hilbert–Pólya files are therefore retained as reusable interfaces and as negative-control examples. They are not merged into the two main routes because they do not supply a non-equivalent analytic lever. Likewise, the 3–4–1 product is a verified classical boundary argument, while the formal certificate barrier records why that family cannot be extrapolated to RH.

4.3 The Jensen literature and the exact role of Route B

The Jensen literature is both relevant and cautionary. The Griffin–Ono–Rolen–Zagier theorem gives Hermite asymptotics and eventual fixed-degree information; the Xi-specific effective work makes the asymptotic regime effective; and O'Sullivan gives a complementary Hermite-combination criterion and detailed coefficient expansions [12, 13, 18]. These results motivate the root-separation and remainder targets in §9.

They do not, however, close Pólya–Jensen. Farmer's analysis argues that the Hermite phenomenon can lose the information needed to attack RH and gives criteria for distinguishing a useful equivalence from an information-losing one [14]. We incorporate that criticism explicitly: Route B is an exact formal receptor, not a claim that Jensen asymptotics are by themselves a plausible proof strategy. The recent finite-strip analysis of Planat further sharpens the distinction between eventual large-shift hyperbolicity and the finite set of low shifts that still carries the RH content [15]. A subsequent experimental study reports parity-dependent onset behavior, finite Hankel positivity, and Padé evidence for a conjectural Stieltjes bridge [16]. We use these observations only to motivate falsification targets; no conjectural bridge is consumed by Route B.

4.4 The PF literature and the discrete/continuous separation

The ASW–Edrei characterization and the Fekete criterion supply the classical total-positivity context [4, 5, 6, 7]. Katkova applies the multiply-positive/PF hierarchy directly to the Riemann Xi generating function and records the relevant consecutive-minor criterion [9]. Our contribution is the Lean proof of the Toeplitz-specific algebraic closure and its exact receptors, not a new classification of PF sequences or discovery of the classical criterion.

A July 2026 preprint gives an explicit uniform tail wedge for the discrete Xi coefficients:

in its normalization, every consecutive minor of order $r \geq 2$ is positive once the shift is at least $10^{18}r^3$ [20]. The result is recent and has not been independently rechecked or transported into Lean here, but it directly informs the revised Route A frontier below. By contrast, the certified PF_5 failure for the continuous de Bruijn–Newman kernel [19] is retained as a route-kill: it prevents silently replacing the discrete factorial-weighted Xi moments by a continuous PF_∞ kernel. Any future proof must establish a discrete transfer theorem or prove the discrete minors directly.

4.5 Position relative to other formalization projects

The Lean Millennium Prize Problems repository formalizes the official problem statements while leaving the RH proof open [25]. Other active Lean projects publish conditional RH reductions with named analytic frontiers [26, 27]. The defensible distinction of the present work is narrower: it provides an axiom-audited Fekete/Toeplitz all-orders closure, an effective-certificate assembly theorem, a bounded-degree hyperbolic-limit theorem, and a Schur–Szegő receptor, together with falsification artifacts for the tempting local routes. We therefore claim a formalized algebraic closure and an exact-ready interface, not the first or only Lean reduction of RH.

5 The Fekete engine

Fekete's classical criterion—strict positivity of solid minors implies strict total positivity [6, 7]—suggests that the contiguous ladder should control all minors. For the one-sided (lower-triangular) Toeplitz setting we formalize a complete, self-contained version of this mechanism. All results in this section are unconditional, kernel-checked theorems.

5.1 A generic Grassmann syzygy and three-term identity

The engine is Xi-free multilinear algebra over an arbitrary commutative ring R (`GrassmannSyzygy.lean`). For $q \geq 0$ let $W\colon \mathrm{Fin}(q{+}3) \to \mathrm{Fin}(q{+}2) \to R$ be a family of $q+3$ row vectors in R^{q+2}, let M_i denote the maximal minor deleting row i, and for $u < v$ let $S(\widehat{u}, \widehat{v})$ denote the order-$(q{+}1)$ minor on the first $q+1$ columns whose rows are the ascending complement of $\{u, v\}$.

Theorem 5.1 (Grassmann syzygy; `maxMinor_syzygy`)**.** *For every column index c, $\sum_{i=0}^{q+2} (-1)^i W_{i,c} M_i = 0$.*

The proof formalizes the classical duplicated-column argument: the $(q{+}3) \times (q{+}3)$ determinant obtained by adjoining the c-th coordinate column to W vanishes, and Laplace expansion along the adjoined column yields the syzygy.

Theorem 5.2 (Three-term identity; `grassmann_three_term`)**.** *For every interior index $0 < t < q+2$,*

$$M_t \cdot S(\widehat{0}, \widehat{q{+}2}) \;=\; M_0 \cdot S(\widehat{t}, \widehat{q{+}2}) \;+\; M_{q+2} \cdot S(\widehat{0}, \widehat{t}).$$

The proof contracts Theorem 5.1 against the cofactor functional of q fixed spectator rows (killing all but three terms by the alternating property) and sorts the surviving subminors into ascending row order, with signs computed by the cycle-range permutation (`Fin.cycleRange`, of sign $(-1)^p$). All signs cancel, yielding the subtraction-free three-term form. At orders 3 and 4 the identity specializes to polynomial identities we had previously verified by `ring`; Theorem 5.2 holds at every order, which is what the induction below consumes.

5.2 The all-orders gap-shrinking induction

Theorem 5.3 (All-orders initial-column theorem; `xiInitialMinor_pos_of_strictLadder`)**.** *Assume the strict contiguous ladder (`XiContigToeplitzStrictPositive`). Then for every k and every strictly increasing row selection $r_0 < \cdots < r_{k-1}$, the initial-column minor* $\det[\,T(r_a, b)\,]_{0\le a,b<k}$ *is strictly positive.*

The proof is a joint induction: an outer induction on the order k (each order divides by a strict order-$(k-1)$ arbitrary-row minor supplied by the previous stage) and an inner strong induction on the *row dispersion* $r_{k-1} - r_0 - (k-1)$. When the dispersion is zero the rows are consecutive and the minor is a ladder rung. Otherwise a row s is inserted into any gap (via `Fin.insertNth`), and Theorem 5.2—applied to the extended $(k+1)$-row family—expresses the target minor, times a strictly positive divisor of order $k-1$, as a nonnegative combination of two order-k minors of strictly smaller dispersion. The two ladder rungs $k = 1, 2$ of the hypothesis are, respectively, strict positivity of the μ_n and the strict Turán inequalities; the formalization records this subsumption (`xiMomentStrictTur an_of_strictLadder`).

5.3 Dominance dichotomy, reversal symmetry, and full PF

Theorem 5.4 (Dominance dichotomy; `xiMinor_eq_zero_of_lt`, `xiMinor_pos_of_dominan t`)**.** *Let $r_0 < \cdots < r_{k-1}$ and $c_0 < \cdots < c_{k-1}$ be arbitrary strictly increasing row and column selections.*

1. *If $r_a < c_a$ for some a, then* $\det[\,T(r_a, c_b)\,] = 0$.
2. *If $c_a \le r_a$ for all a and the strict contiguous ladder holds, then* $\det[\,T(r_a, c_b)\,] > 0$.

Part (1) is a permutation-pigeonhole argument on the lower-triangular zero pattern: every permutation term contains an entry above the support. Part (2) repeats the gap-shrinking induction with the columns fixed but arbitrary; its base case (consecutive rows, arbitrary dominated columns) reduces to Theorem 5.3 by a Toeplitz *reversal symmetry* (`x iMinor_contigRows`): reversing both index orders and transposing carries a contiguous-row minor to an arbitrary-row initial-column minor. In the induction step the only possibly degenerate term is the drop-last-row flanking minor, and there part (1) supplies the required nonnegativity.

Corollary 5.5 (Full PF from the strict ladder; `xiToeplitzTotalPositive_of_strictLadd er`)**.** *The strict contiguous ladder implies the full Pólya frequency condition `XiToeplitzTo talPositive`.*

Corollary 5.6 (Sharpest reduction, and RH-strength labeling; `riemannHypothesis_of_str ictLadder`)**.** *Given the classical scaffolding, the strict contiguous ladder implies RH. Since, conversely, the scaffolding makes the full PF condition* equivalent *to RH (`xiToeplitzTota lPositive_iff_riemannHypothesis`), the strict ladder is at least RH-strength: it can never be assumed, axiomatized, or verified numerically.*

Corollary 5.5 eliminates two inputs that earlier versions of the reduction required: Cryer's lower-triangular initial-column criterion [8] and Pólya's kernel representation are no longer consumed by the minimal path (both remain available, as named classical inputs, to secondary routes).

6 Route A: effective kernel certificates

Under Pólya's representation the coefficient-side and kernel-side contiguous minors are related by the exact scaling $D_k(m) = 2^k \cdot D_k^{\mathrm{ker}}(m)$, where D_k^{ker} is built from the factorial-weighted moments $M_n/(2n)!$; strict positivity therefore transfers verbatim (`xiToeplitzContigMinor_pos_iff_moment_of_kernelRep`). This suggests organizing the open mathematics *order by order*:

Definition 6.1 (Per-order effective certificates; `KernelEffectiveCertificates`)**.** For a moment model M: for every order k there exists a cutoff N with

- *certified prefix* (`KernelMinorPrefixStrict`): $D_k^{\mathrm{ker}}(m) > 0$ for all $m < N$ — for rational interval enclosures of the moments this is dischargeable offset-by-offset by `norm_num`;
- *explicit remainder tail* (`KernelMinorTailStrict`): $D_k^{\mathrm{ker}}(m) > 0$ for all $m \geq N$ — the target of effective asymptotic analysis.

Theorem 6.2 (Route A; `riemannHypothesis_of_effectiveCertificates`)**.** *Classical scaffolding + Pólya kernel data + `KernelEffectiveCertificates` $\Longrightarrow$ RH.*

Every interface in this statement is a proved theorem: certificates assemble into the kernel-side strict ladder, transfer to the coefficient-side ladder, and feed Corollary 5.6. The order-3 instance connects to an existing prefix/ratio-tail decomposition of the first open rung (compatibility lemma `order3TailGap_of_kernelTailStrict`); the current interval scans certify the contiguous order-3 prefix through offset 116 with smallest normalized value $\approx 4.2 \times 10^{-6}$. In the corrected row box $0 \leq r_i \leq 48$, all 18,424 order-3 and 211,876 order-4 arbitrary-row initial-column selections are certified positive, with no negative, zero, or inconclusive intervals. The complete row-48 scan is now reproduced independently by Python and TypeScript exact interval implementations and a JavaScript midpoint implementation; the row-30 baseline remains available for regression comparison. These are falsification evidence, not proof.

7 Route B: the Jensen/Schur–Szegő receptor

The second route runs through Pólya's equivalence of RH with hyperbolicity (real-rootedness) of all Jensen polynomials $J_{d,n}(x) = \sum_{k \leq d} \binom{d}{k} b_{n+k} x^k$ [3].

Theorem 7.1 (Bounded-degree hyperbolic limits; `polynomialHyperbolic_of_tendsto`)**.** *Let p_j be real polynomials of degree at most D, each hyperbolic, whose complexifications converge pointwise on $\mathbb{C}$ to that of a nonzero real polynomial f. Then f is hyperbolic.*

The formal proof is elementary and complex-analysis-free: for a real-rooted polynomial and two points on a horizontal line $\operatorname{im} z = y \neq 0$, each root factor obeys $|z_2 - r|^2 \leq K|z_1 - r|^2$ with $K = 2 + 2(\operatorname{re} z_2 - \operatorname{re} z_1)^2/y^2$, so a nonreal zero of the limit would force vanishing along the entire line. This is a proved leg of the Laguerre–Pólya closure bridge, which the formalization correspondingly reduces to a sharper named condition (`XiJensenApproximationBridge`).

The Jensen polynomial is exactly the Schur–Szegő binomial composition of its coefficient-window polynomial with $(1 + X)^d$ (`JensenPoly_eq_schurSzego_one_add_X_pow`), and the general same-sign composition theorem [21, 22] enters as the named target `HermitePoulainSchurSzegoTheorem`; its low-degree cases ($d \leq 3$), the hyperbolicity-preservation of differentiation, and a counterexample showing the *unrestricted* composition claim is false are proved in the repository. The receptor result discharges the fixed factor's side conditions once and for all:

Theorem 7.2 (Schur–Szegő receptor; `jensenPoly_hyperbolic_of_hpss`, `riemannHypothesis_of_hpss_route`)**.** *The polynomial* $(1+X)^d$ *has degree* d*, is hyperbolic, and has all real roots of one sign. Consequently, given* (i) *the Pólya–Jensen bridge,* (ii) *the general same-sign Schur–Szegő theorem, and* (iii) *hyperbolicity with full degree of every coefficient-window polynomial of* (b_n) *(`XiJensenCoeffHyperbolic`), RH follows.*

8 Machine-checked falsification results

Theorem 8.1 (Local Turán data cannot close order 3; `turan23_not_imp_toeplitz3`)**.** *There exist positive reals* $b_0, \dots, b_4$ *satisfying the strict second-order Turán inequalities and both third-order Turán inequalities on the window, for which the contiguous* 3×3 *Toeplitz determinant is negative. A witness is* $(b_0, \dots, b_4) = (\frac{1}{100}, 1, \frac{5}{4}, 1, \frac{1}{100})$*, with Turán margins* ≈ 0.68 *and determinant* $\frac{45}{64} - \frac{31}{25} + \frac{79}{8000} \approx -0.53$*.*

Since the second-order inequalities are the CNV theorem [10] and the third-order inequalities are the Dimitrov–Lucas theorem [11], Theorem 8.1 shows that the two strongest known unconditional local results about (μ_n) cannot, by sequence-level algebra alone, prove even the first open rung of the ladder: any proof must use quantitative information that the five-term Turán window forgets (the Ξ windows have $\mu_m^2/(\mu_{m-1}\mu_{m+1}) \to 1^+$ with specific super-exponential decay, far from the witness's regime).

Figure 2 shows that the witness is not an isolated accident. Among about 9.1×10^5 sampled five-term windows that satisfy every order-2 and order-3 Turán inequality, about 2.0×10^5 have $D_3 < 0$. Negative determinants occur at every margin scale, including the small margins 0.008–0.29 occupied by the Ξ windows; in the margin band 10^{-4}–10^{-2} they are 43% of the samples. The sign of D_3 is therefore information that the Turán data do not carry.

Separately, a recent computer-certified failure of PF_5 for the continuous de Bruijn–Newman translation kernel $\Phi(|u|)$ [19] rules out any argument that silently upgrades that kernel to PF_∞; it does *not* bear on the discrete moment sequence targets above, and the repository records this separation so the two objects cannot be conflated.

9 The new mathematics required for an RH proof

The formal reduction is now short enough that the remaining mathematics can be stated as explicit proof obligations rather than as a general request for "more asymptotics." This section gives the proposed statements in the notation consumed by the Lean receptors. A statement marked *target* is not claimed as a theorem of the present repository; it is a precise mathematical problem whose proof would close a named interface. The first two conditional capstones below are proved in Lean once their hypotheses are supplied; the heat-flow item is a proposed third receptor.

9.1 The order-three tail as an interval theorem

For a positive coefficient sequence (μ_n), write

$$\rho_n := \frac{\mu_{n+1}}{\mu_n}, \qquad \Delta_3(m) := \det \begin{bmatrix} \mu_m & \mu_{m-1} & \mu_{m-2} \\ \mu_{m+1} & \mu_m & \mu_{m-1} \\ \mu_{m+2} & \mu_{m+1} & \mu_m \end{bmatrix}, \qquad G_3(m) := \frac{\Delta_3(m)}{\mu_m^3}. \tag{2}$$

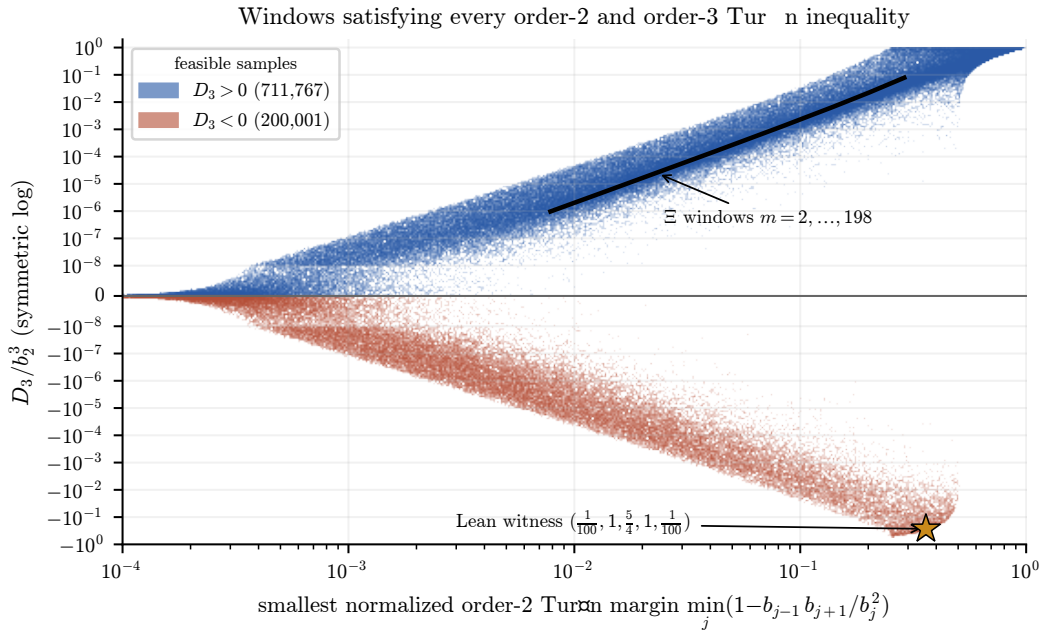


Figure 2: Five-term windows $(b_0, \ldots, b_4)$ obeying every strict order-2 Turán inequality and both order-3 (Dimitrov–Lucas) inequalities, normalized by b_2. Horizontal axis: the smallest relative order-2 margin. Vertical axis: the contiguous determinant D_3/b_2^3 on a symmetric logarithmic scale. Blue and red: random samples with $D_3 > 0$ and $D_3 < 0$ (log-uniform windows plus perturbed log-quadratic windows; seed 20261010). Black: the Ξ windows $m = 2, \ldots, 198$, from midpoints of the certified coefficient enclosures. Star: the Lean witness of Theorem 8.1. Generator: `paper/generate_paper_figures.py`.

For $m \geq 2$, direct division by μ_m^3 gives the ratio expression

$$G_3(m) = 1 - 2\frac{\rho_m}{\rho_{m-1}} + \frac{\rho_m^2}{\rho_{m-2}\rho_{m-1}} + \frac{\rho_m\rho_{m+1}}{\rho_{m-1}^2} - \frac{\rho_m\rho_{m+1}}{\rho_{m-2}\rho_{m-1}}. \tag{3}$$

This identity is useful because asymptotic results naturally control ratios, whereas the Lean target is a determinant.

Definition 9.1 (Effective ratio box)**.** An effective ratio box at offset m is a tuple of rational intervals

$$I_{m,j} = [\underline{\rho}_{m+j}, \overline{\rho}_{m+j}] \quad (j = -2, -1, 0, 1), \qquad 0 < \underline{\rho}_{m+j} \leq \overline{\rho}_{m+j}, \tag{4}$$

containing the four adjacent ratios. Let $\mathcal{G}_3(I_{m,-2}, \ldots, I_{m,1})$ be the natural interval evaluation of the right-hand side of (3).

Named Target 9.2 (Effective $k = 3$ remainder compiler)**.** Suppose that, for some explicit N, effective asymptotics produce ratio boxes (4) for every $m \geq N$ and that

$$\inf \mathcal{G}_3(I_{m,-2}, I_{m,-1}, I_{m,0}, I_{m,1}) > 0 \qquad (m \geq N). \tag{5}$$

Then `KernelMinorTailStrict`$(M, 3, N)$ holds. In particular, the proof is an exact rational interval calculation once the asymptotic error terms have been translated into the four intervals.

The target isolates the genuinely new analytic step: one must obtain an error margin that survives the cancellation in (3). An asymptotic equivalence without an explicit signed remainder is insufficient. O'Sullivan's Lambert-W expansions [18] are a natural source of such bounds; the missing work is the remainder translation, not another Lean interface.

Figure 3 shows how small the order-3 margin already is. On the certified range the normalized gap $G_3(m)$ decays roughly like $m^{-2.8}$, falling to 4.2×10^{-6} at the end of the certified prefix. An effective ratio box must therefore resolve the four ratios to better than this relative scale, uniformly in m.

9.2 From one rung to every order

The Fekete engine gives the following exact capstone, which is the central all-order theorem of the formalization.

Theorem 9.3 (All-order positivity capstone)**.** *Assume the classical Toeplitz scaffolding and*

$$\forall k, m \in \mathbb{N}, \qquad D_k(m) > 0. \tag{6}$$

Then every minor of the signed Xi Toeplitz matrix is nonnegative and RH holds. In Lean this is `riemannHypothesis_of_strictLadder`*, obtained through* `xiToeplitzTotalPositive_of_strictLadder`*.*

The theorem is conditional because (6) is itself at least RH-strength. Route A packages the order-by-order version of this obligation: a finite certified prefix plus a tail certificate for each k. Thus an all-order asymptotic theorem need not mention Toeplitz minors at all; it only has to imply the named proposition `KernelEffectiveCertificates`.

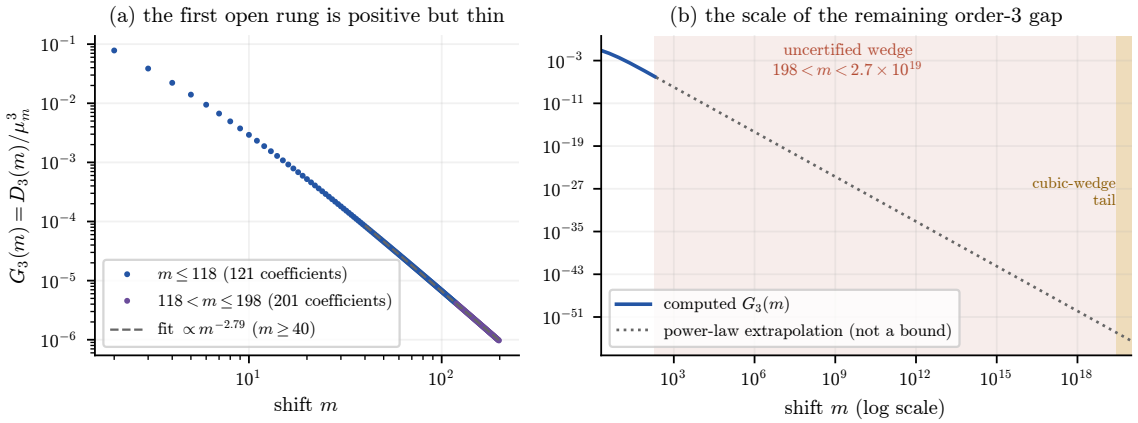


Figure 3: The first open rung. (a) $G_3(m) = D_3(m)/\mu_m^3$ from midpoints of the rigorous coefficient enclosures; all values are positive. Blue: the 121-coefficient prefix used in Route A; purple: the extension to 201 coefficients. The least-squares slope is fitted on $m \ge 40$. (b) The same data on the full scale up to the external cubic-wedge cutoff 2.7×10^{19} [20]; the dotted extrapolation only indicates scale and is not a bound. The rose band is the uncertified order-3 wedge.

The order-three compiler above is the first nontrivial instance. A recent preprint now supplies a much stronger coefficient-side statement [20]: for every order $k \ge 2$,

$$D_k(m) > 0 \qquad \text{whenever} \qquad m \ge 10^{18}k^3. \tag{7}$$

Its matrix is the transpose of our consecutive block. In the preprint's normalization $G(z) = \frac{1}{8}\xi(\frac{1}{2} + \frac{1}{2}\sqrt{z})$, its positive coefficient is $a_n = \mu_n/(8 \cdot 4^n)$; hence an order-$k$ minor at shift m is multiplied by the positive factor $8^{-k}4^{-km}$, so the determinant signs agree. We have not independently rerun the preprint's Arb ancillary certificates or formalized this transport; (7) is therefore cited as an external preprint theorem, not counted among the 122 audited Lean theorems.

If (7) is accepted, the former uniform-tail target has an explicit solution $N(k) = 10^{18}k^3$. The remaining Route A obligation is the complementary wedge.

Named Target 9.4 (Complementary finite-prefix wedge)**.** Certify $D_k(m) > 0$ for every $k \ge 2$ and every $0 \le m < 10^{18}k^3$, in the exact coefficient normalization consumed by the Lean receptor. Together with (7), these certificates instantiate the strict contiguous ladder.

The word "finite" is per order only: the union over all k remains infinite and RH-strength. For order three alone the external cutoff is 2.7×10^{19}, so the current prefix through offset 116 leaves an enormous interval. The preprint changes the shape of the frontier from an unknown tail to a certified-tail/uncertified-wedge split; it does not make the remaining verification computationally routine.

Figure 4 draws this split in the order–shift plane.

9.3 A root-separation theorem for the Jensen route

Let $p(x) = a\prod_{i=1}^{d}(x - r_i)$ be a real polynomial with distinct real roots, and choose $0 < s < \frac{1}{2}\min_{i\neq j}|r_i - r_j|$. Set

$$\eta(p, s) := \min_{1\le i\le d}\ \min_{|z-r_i|=s} |p(z)| > 0. \tag{8}$$

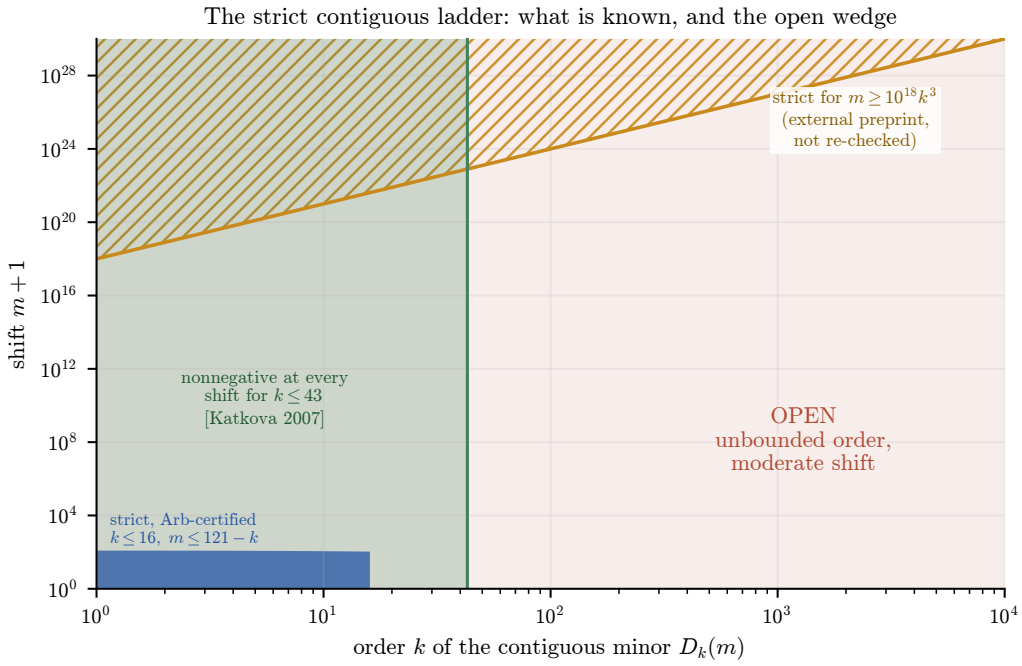


Figure 4: Where the contiguous determinants $D_k(m)$ are controlled, on logarithmic axes. Blue: strict positivity certified by exact interval arithmetic ($k \leq 16$, $m \leq 121-k$). Green: nonnegativity of every minor of order at most 43 at every shift [9]; this gives ≥ 0, not the strict inequality the Fekete engine consumes. Hatched: the external cubic wedge $m \geq 10^{18}k^3$ [20], not re-checked here. Rose: the complementary wedge of unbounded order and moderate shift, where the open content of Route A lies.

Theorem 9.5 (Root-separation perturbation)**.** *Let q be a real polynomial of degree d and suppose*

$$\sup_{|z-r_i|=s} |q(z)-p(z)| < \eta(p,s) \qquad (1 \leq i \leq d). \tag{9}$$

Then q has exactly one zero in each disk $|z-r_i| < s$, and every zero of q is real. Consequently q is hyperbolic.

Indeed, Rouché's theorem gives one zero in each disk. The disks are pairwise disjoint and q has degree d, so these are all its zeros. A nonreal zero in a disk centered on the real axis would bring its conjugate into the same disk, contradicting the count one. This theorem is elementary but exact-ready: coefficient bounds imply the left side of (9) by the triangle inequality.

Named Target 9.6 (Xi Jensen window certificate)**.** For every degree d there are explicit $n_0(d)$, Hermite-model polynomials H_d, and margins $s_d, \eta_d > 0$ such that for every $n \geq n_0(d)$ the normalized coefficient-window polynomial of the Xi coefficients satisfies (9) with $p = H_d$. The finite set of windows $n < n_0(d)$ is discharged by exact algebra or certified interval arithmetic. This target implies `XiJensenCoeffHyperbolic`.

The asymptotic input from Griffin–Ono–Rolen–Zagier [12] gives the Hermite shape; an effective version must additionally bound the root separation margin and the coefficient error in a common normalization. This is a concrete alternative to proving the full same-sign Hermite–Poulain/Schur–Szegő theorem for every degree.

9.4 Conditional capstones and a possible heat-flow route

The exact-ready architecture can be summarized as the following theorem schema. It is deliberately stated with named propositions, so the proof does not hide a conjecture in a typeclass or an imported declaration.

Theorem 9.7 (Three conditional RH capstones)**.** *Assume the classical analytic scaffolding. Each of the following hypotheses is separately sufficient for RH:*

1. `KernelEffectiveCertificates`;
2. `PolyaJensenBridge`, `HermitePoulainSchurSzegoTheorem`, *and* `XiJensenCoeffHyperbolic`;
3. *a proved heat-flow rigidity proposition* `HeatFlowRigidity` *asserting that the de Bruijn–Newman constant satisfies* $\Lambda \le 0$.

The first two implications are the Lean theorems `riemannHypothesis_of_effectiveCertificates` *and* `riemannHypothesis_of_hpss_route`*; the third combines* $\Lambda \ge 0$ *of Rodgers–Tao [17] with the classical equivalence* $\mathrm{RH} \iff \Lambda \le 0$.

The third item is a research direction, not a hidden assumption. A possible mathematical target is a quantitative backward-heat-flow rigidity statement: if $\Lambda > 0$, then a specified zero-statistics inequality is violated at an explicit height. Proving that implication would produce `HeatFlowRigidity` and close a third route without Toeplitz positivity.

9.5 What remains genuinely open

The four classical bridges (ASW/Edrei, Laguerre–Pólya closure, Pólya–Jensen, and the converse) are formalization work, albeit substantial work involving entire functions and Hadamard products. Subject to independent verification, the cubic-wedge preprint supplies the all-order tail shape formerly missing from Route A; what remains is its normalization-exact Lean transport and the complementary finite-prefix wedge. The Jensen route still requires an effective root-separation certificate such as (9). None of these remaining statements is currently proved in the repository in the exact Lean form.

10 Discussion and conclusion

Formalized analytic number theory has advanced rapidly—Mathlib contains the nonvanishing of ζ on $\mathrm{Re}(s) \ge 1$, the functional equation, and discreteness of the zero set, while independent Lean projects now formalize other RH reductions and the official Millennium-problem statement [25, 26, 27]. Our narrower claim is that the present repository combines a generic Grassmann three-term gap-shrinking identity, a Toeplitz-specific all-orders closure, and two axiom-audited exact-ready receptors in one checked development. The counterexample of Theorem 8.1 appears to be new as a stated separation, though it is elementary once asked.

The honest summary is unchanged from the abstract: RH is not proved, and the remaining inputs are provably at least as hard as RH (the strict ladder) or are classical formalization debt (the scaffolding). What has changed is the shape of the problem: the open mathematics is now a short list of named statements, each with a kernel-checked theorem waiting to consume it. All results are reproducible from the repository by running `lakebuild`, the axiom-audit script, and the verification harness; the audit reports 122 theorems depending only on Lean's foundational axioms.

A Reproducibility and theorem-status ledger

This release has not received an independent human expert report. The review questions and evidence bundle are recorded in `research/INDEPENDENT_REVIEW_PACKET_2026_07_17.md`; kernel checking and a literature audit should not be represented as external mathematical certification.

The Lean baseline used for this revision is Lean 4.30.0 ("`leanprover/lean4:v4.30.0`") with Mathlib revision `c5ea00351c28e24afc9f0f84379aa41082b1188f`. The repository baseline commit preceding this revision is `c4729cd`. The release archive deposited with this version (doi:10.5281/zenodo.23187646; all versions: doi:10.5281/zenodo.21416394) records the final commit that includes this manuscript and the scan reports. The following commands were run successfully:

```
lake build Reinmann
bash scripts/verify_axiom_clean.sh
bash scripts/run_scan_48.sh
python3 -m pytest -q \
  experiments/cross_language_toeplitz/test_uncertainty.py
bash scripts/run_cross_language_toeplitz.sh 30
bash scripts/run_cross_language_toeplitz.sh 48 \\
  research/cross_language_toeplitz/row48
```

The first command completed 3527 jobs. The axiom audit reported 122 headline theorems depending only on `[propext, Classical.choice, Quot.sound]`; the source scan found no `sorry`, `admit`, or project-declared axioms. The shortcut and full-build phases of `safe-verify.sh` also passed over 72 Lean files; its redundant per-file recompilation phase was sampled without failures but was not rerun to completion for this paper/script-only revision and remains a final submission gate. The row-48 numerical scans reported all 18,424 order-three and 211,876 order-four initial-column minors as certified positive, with zero negatives or inconclusive intervals. The cross-language row-48 run completed in 453.26 seconds and produced matching sign-stream hashes across the two interval implementations and the JavaScript midpoint check. This run uses an exponent-aware half-ULP parser; an earlier parser ignored the exponent when assigning uncertainty to scientific-notation tokens and therefore produced invalidly wide intervals. Regression tests cover that defect. The coefficient cache remains provenance for a numerical diagnostic, not a formal proof of the Xi coefficients.

The status ledger is:

- *Proved in Lean:* the Xi target-function correction, the generic Grassmann syzygy and three-term identity, the all-orders Fekete closure, the bounded-degree hyperbolic-limit theorem, the Turán route-kill, and the two conditional assembly theorems.
- *External classical input:* the Pólya kernel representation, ASW/Edrei, Laguerre–Pólya closure, and Pólya–Jensen bridge.
- *External recent preprint theorem:* the cubic coefficient-tail wedge (7). Its normalization transport and computer-assisted certificates have not been checked by Lean or independently rerun in this repository.
- *Open analytic targets:* `KernelEffectiveCertificates`, `HermitePoulainSchurSzegoTheorem`, `XiJensenCoeffHyperbolic`, and the proposed heat-flow rigidity proposition. None is imported as an axiom.

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